

Physical Optics Sim - Documentation

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Purpose

This code implements fresnel and fraunhofer diffraction techniques, as well as beam steering to accurately incorporate all aspects of the BICEP Array calibration process. Replicating this process can allow us to compare the data from the telescope beam maps to the simulation and ultimately finding a quantitative assessment of how the mirror non-flatness might be affecting areas of the beam maps.

Global Variables

Global variables allow for customization of the simulation. The variables are as follows:

Variable	What it does
N	number of grid points
Radius	Radius of circular aperture
dx	spacing between grid points
edge taper	controls the gaussian fade
theta 1 and theta 2	Steering angles for tilt adjustments
lamda	wavelength of light being observed
mirror coord	Cartesian coordinates for the mirror system
window size	Size of the grid used in Fresnel diffraction
phase gradient	.
z far	Far-field distance in ()

Table 1: Available variables

Aperture Creation

The aperture function focuses on creating a 2d spatial grid that defines a circular aperture. It uses an N number of grid points along each axis, and grid spacing (dx) in the x and y directions. The aperture is generated with a Gaussian edge taper to simulate real techniques. This means that some of the parts of the grid are non-zero, simulating a circular aperture. The function used to help create this is as follows:

$$r = \sqrt{x[i,j]^2 + y[i,j]^2}$$

The dx remains the same for the aperture function and the fresnel function, since both use the same physical grid spacing. (1)

The coordinate array is centered at zero and a gaussian taper is applied to replicate the illumination of the aperture in experimental values.

BEAM STEERING

$$\phi = k(x\cos(\theta) + y\sin(\theta))\tan(\alpha)$$

In order to steer the beam source two different angles must be defined. Here, there is a tilt angle (

α

) on the x-axis. (insert ref)

alpha steer is the tilt angle. This describes the angle the beam makes in the z direction.

theta steer is the azimuth angle, which is the xy direction that the beam is leaning in.

spatial frequencies and slope can then be defined using trigonometry.

When both angles are zero, the tilt phase is zero, meaning that the beam is pointing straight ahead.

If you want to steer along the x or the y axis (right or left) the theta steer angle can be changed to achieve this.

For example, when wanting to move along the x axis theta should remain at zero and the amount you'd like to travel along the x axis can be changed using alpha steer.

For the y axis direction, change theta steer to 90 degrees and then use the alpha steer to determine how far from center along the y axis you'd like to steer.

This method can then be applied to different scenarios to achieve different directional steering.

Mirror Creation

The mirror function creates new coordinates that can be used for the fresnel calculation. It also implements beam steering (theta) for tilt adjustments of the beam.

DISTORTION:

Fresnel Diffraction

This function uses fresnel techniques as found in (ref Optical physics third edition) to calculate the diffraction pattern from the aperture onto the mirror. Rather than using an FFT here, a for loop is utilized with the diffraction by Huygen's principle:

$$E(x', y') = \iint A(x, y) \frac{e^{ikd}}{d} dx dy$$

The for loop ensures that the Fresnel diffraction technique is preserved and accurate in the near field.

This function also accounts for the complex phase shift that is created from the tilted mirror

ACCOMODATING THE ANGLED REFLECTION: (multiply by -1 terminology) Since the Fraunhofer calculation is simulated as it reflects off the mirror at a 45 degree angle, the Near field output needs to be multiplied by -1 to account for this change in direction. According to Euler's formula, we can multiply the following phase shift by the produced output to achieve this effect.

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1 + 0j = -1$$

Fraunhofer Diffraction

A two step propagation: When doing Fraunhofer diffraction from the plane of the mirror to the far field, an FFT can be used.

Both a log scale and the linear scale of the FFTs are visible below:

The nearfield output from the fresnel diffraction function is input into the fraunhofer function. This allows for all features applied to be carried through accurately into the far field projection.

Fresnel diffraction integral calculation using a single FFT method explicitly includes the distance and quadratic phase factor to output a grid in physical mm of distance. To go from this output to an angular deviation, the following equation can be used:

$$\tan(\theta_x) = \frac{x_2}{z_1}$$

Where z_1 is the Far Field propagation distance. This formula yields an angle in radians which can then be converted to degrees for plot interpretation.

When considering the uneven axis on the mirror and how that's handled in the FFT we have to keep in mind the inverse relation with spatial dimensions.

$$f_x \propto \frac{1}{\delta x}$$

Since the mirror coordinates are stretched along the x-axis the FFT pattern needs to look compressed on the x-axis.

Hygens principle for explanation of variable c FOURIER SHIFT THEROEM:

Have to try beam tracking, to find out where the fft is projecting to, and then center the frame around that.

FIXING THE RADICAL TWO Y AXIS COMPRESSION

Optical propogation law:

$$\Delta_{x_{out}} = \frac{\lambda_z}{L_x} \quad (1)$$

Where L_x is equal to N_x times $\Delta_{x_{in}}$.

$$\Delta_{y_{out}} = \frac{\lambda_z}{L_y} \quad (2)$$

Where L_y is equal to N_y times $\Delta_{y_{in}}$.

When the intermediate aperture of the near field mirror is used, the FFT calculation compresses the image in the y axis direction. This can be adjusted in the plotting steps when the y max and mins are calculated for the FFT plot. An adjustment factor of root two can help to recenter the image.

Scalar diffraction theory

- Peak Normalization - Fraunhofer function: <https://blogs.bath.ac.uk/python/finding-maximum-values-from-arrays/>
- Diffraction Scaling - Voelz, David George, and Society of Photo-optical Instrumentation Engineers. Computational Fourier Optics a MATLAB Tutorial. SPIE, 2011, <https://doi.org/10.1117/3.858456>.
- Goodman, Joseph W. Introduction to Fourier Optics. McGraw-Hill, 1968.